

Post Graduate Diploma in Cyber Security and Law

Semester II — End Semester Examination (2025–26)

Paper No. 4 | Paper Code: 92911203 | Subject: Cyber Law & Forensics

GUESS PAPER / IMPORTANT QUESTION BANK

UNIT 1: Introduction to Cyber Law & the IT Act, 2000

1. Define Cybercrime and list any four examples of it.
2. Describe the primary objectives and coverage of the Information Technology Act, 2000.
3. Define Cyber Defamation and explain how it differs from conventional defamation, citing one real-life instance.
4. Analyse the idea of Cyber Espionage and how it differs from Cyber Terrorism, using documented real-world cases as support.
5. Evaluate the importance of E-Commerce for modern business and the legal risks that come with online transactions.
6. Discuss how personal data is misused on social media platforms, and assess whether the IT Act, 2000 and the Digital Personal Data Protection Act, 2023 adequately address this misuse.

UNIT 2: Digital Signatures, Certifying Authorities & E-Governance

7. Differentiate between a Digital Signature and an Electronic Signature, and state the conditions an Electronic Signature must meet to be treated as 'Reliable' under Section 3A of the IT Act, 2000.
8. Define Digital Signature and describe, step by step, how it functions. Why does Indian law treat it as legally valid?
9. Who qualifies as a Certifying Authority (CA) under the IT Act, 2000? Outline its key responsibilities and the step-by-step process for issuing a Digital Signature Certificate (DSC).
10. Define Electronic Governance (e-Governance) and give four real-life examples of e-Governance services offered in India.
11. Define Electronic Evidence and describe the conditions under which electronic records become admissible in court as per Section 65B of the Indian Evidence Act.
12. Describe the concept of an Electronic Record under the IT Act, 2000, along with its legal standing.

UNIT 3: Cyber Offences, Security Policies & Regulatory Bodies

- 13.** Write brief notes on any three cyber offences defined under the IT Act, 2000, stating the relevant section, the nature of the offence, and its punishment for each.
- 14.** Define CERT-In and describe its key functions under Section 70B of the IT Act. Why does it matter for India's cybersecurity landscape?
- 15.** Define Security Policy and explain why organisations need one. Describe any four categories of security policies with an example each, and give a short account of the CIA Triad.
- 16.** Describe the 'Safe Harbour' protection extended to intermediaries under Section 79 of the IT Act.
- 17.** Discuss the Central Government's authority to block online content under Section 69A of the IT Act, 2000.
- 18.** Describe the offences covered under Sections 66, 66C, 66D, and 67 of the IT Act, 2000, using suitable examples.
- 19.** Define Cyber Terrorism as covered under Section 66F of the IT Act, 2000, and explain how it differs from regular hacking.
- 20.** Describe Phishing and Identity Theft as forms of cybercrime, and outline the legal provisions in the IT Act that address them.

UNIT 4: Data Protection & Privacy Law

- 21.** Define GDPR and describe its core principles and the rights it grants individuals. State the penalties for GDPR violations and explain its relevance to Indian companies.
- 22.** Describe the key features of the Digital Personal Data Protection Act, 2023.
- 23.** Discuss the Right to Privacy as a fundamental right in India, referring to the Justice K.S. Puttaswamy judgment, and its bearing on cyber law.
- 24.** Compare the GDPR with the Digital Personal Data Protection Act, 2023, on the basis of scope, consent requirements, and enforcement mechanisms.
- 25.** Describe the role and powers vested in the Data Protection Board of India under the DPDP Act, 2023.

UNIT 5: Cyber Forensics Fundamentals & File Systems

- 26.** Define Cyber Forensics and explain its importance, describing the standard steps followed during a cyber forensic investigation in proper sequence.
- 27.** Describe the idea of Storage Fundamentals in cyber forensics and explain why file system structures (FAT32, NTFS, ext4) matter for recovering digital evidence.
- 28.** Define File System and give a brief account of FAT32 and NTFS. How do these two file systems treat deleted files, and why does this matter to a forensic investigator?
- 29.** Describe the categories of digital evidence — volatile and non-volatile — giving examples of each.

30. Discuss why metadata matters in a digital forensic investigation, giving examples of metadata that investigators find useful.

31. Describe what a write blocker is and how it helps preserve the integrity of digital evidence during acquisition.

UNIT 6: Evidence Handling — Chain of Custody & Forensic Imaging

32. Define Chain of Custody and describe, step by step, the complete process of maintaining it during a cybercrime investigation, from evidence collection at the scene through to its presentation in court. Why can a broken chain of custody prove fatal to a case?

33. What civil and criminal liability can a person or organisation face under Sections 43 and 66 of the IT Act, 2000, for unauthorised access to or damage of a computer system?

34. Define forensic image and explain how it differs from an ordinary copy of data. Name the tools generally used to create a forensic image, and describe how hash values help verify its integrity.

35. Explain how MD5 and SHA-256 hashing algorithms differ when it comes to establishing the integrity of digital evidence.

36. Discuss the legal and procedural requirements for the Search and Seizure of computers under Indian law, and the precautions an investigator must observe so the evidence stays admissible in court.

37. Describe the concept of Anti-Forensics and the common methods offenders use to defeat or mislead a forensic investigation.

UNIT 7: Data & Artefact Recovery

38. Define Deleted File Recovery and explain how deleting a file is different from wiping it permanently. What forensic tools and techniques help recover a file from a storage device?

39. What does Recovering Deleted Evidence mean? Name any two forensic tools used for this and briefly describe how each one functions.

40. Define Temporary Files, Cache Files, and Swap Files, giving a simple example of each, and explain how they assist a forensic investigator.

41. Discuss how internet browsing history and browser artefacts (cookies, cache files, download history, etc.) assist a cyber forensic investigation. Name two forensic tools used to retrieve browser history.

42. Discuss the value of the Windows Registry as an evidence source during a forensic investigation.

43. Define Data Carving and explain how it enables file recovery even when file system metadata is missing or damaged.

UNIT 8: Email, Network, Mobile & Password Forensics

44. Describe the process of Email Investigation in cyber forensics. What key details are found in email headers, and how do investigators use them to trace the origin of a suspicious or criminal email?

45. Discuss the role of Password Cracking methods — such as Dictionary Attacks, Brute Force, and GPU-accelerated cracking with Hashcat — in cyber forensic investigations.
46. Define Network Forensics and describe how log files and packet-capture tools (e.g., Wireshark) assist in investigating a network-based cyberattack.
47. Define Mobile Device Forensics and discuss the challenges of extracting and preserving evidence from smartphones.
48. Define Cloud Forensics and discuss the jurisdictional and technical hurdles involved in investigating evidence held on cloud platforms.
49. Discuss the role and main features of any two forensic tools or software (e.g., EnCase, FTK, Autopsy, Cellebrite) used in digital investigations.
50. Write a brief note on Steganography and its role in hiding as well as detecting digital evidence.

UNIT 9: Emerging & Miscellaneous Topics

51. Discuss the jurisdictional challenges involved in investigating and prosecuting cross-border cybercrime.
52. Explain the key Intellectual Property Rights (IPR) issues that arise in cyberspace, such as online copyright and trademark infringement.
53. Define Cyberstalking and Cyberbullying, and discuss the legal provisions available in India to address them.
54. Discuss the offence of publishing or transmitting child sexual abuse material under Sections 67A and 67B of the IT Act, 2000.
55. Are contracts formed electronically (e-contracts) legally valid in India? Discuss the essential conditions for their validity under the IT Act, 2000, and the Indian Contract Act.
56. Explain the 'Order of Volatility' in digital forensics and why it determines the sequence of evidence collection during a live investigation.
57. Discuss the significance of RAM (memory) forensics and describe the type of evidence that can be recovered from volatile memory.
58. Explain the role of Windows Event Logs in a forensic investigation and the kind of information they record.
59. Write a short note on India's National Cyber Security Policy and its key objectives.
60. What is meant by Digital Forensic Readiness? Discuss its importance for organisations in responding effectively to a security incident.

Total Questions Compiled: 60
